MIL-T-7258B(USAF)
20 APR 85
SUPERSEDING
MIL-B-7258A(USAF)
16 November 1970

MILITARY SPECIFICATION

TRANSMITTER, TEMPERATURE, ELECTRICAL RESISTANCE, TYPE B-4

This specification is approved for use by the Department of the Air Force, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE.

* 1.1 Scope. This specification covers one type of electrical resistance free air temperature transmitter, range -70 to ±70° Centigrade (C) designated Type B-4.

2. APPLICABLE DOCUMENTS

* 2.1 Government documents.

* 2.1.1 Specifications and standards. Unless otherwise specified (see 6.2), the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-B-566	Box, Folding Paperboard
PPP-B-636	Box, Shipping, Fiberboard

MILITARY

MIL-P-116	Preservation, Methods Of
DOD-D-1000	Drawing, Engineering And Associated List
MIL-R-6855	Rubber, Synthetic, Sheet, Strips, Molded Or Extruded Shapes
MIL-S-7742	Screw Threads, Standard, Optimum Selected Series, General Specification For

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: the Engineering Division, San Antonio ALC/MMEDO, Kelly AFB, Texas 78241 by using the self addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 6685

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STANDARDS

MILITARY

	MIL-STD-105	Sampling Procedures And Tables For Inspection by Attributes
	MIL-STD-129	Marking For Shipment And Storage
	MIL-STD-130	Identification Marking Of US Military Property
	MIL-STD-143	Standards And Specifications, Order Of Precedence For The Selection Of
*	MIL-STD-794	Part And Equipment, Procedures For Packaging And Packing Of
*	MIL-STD-810	Environmental Test Methods And Engineering Guidelines
	MS24693	Screw, Machine, Flat Countersunk Head, 100°, Cross Recessed, UNC-2A And UNF-2A
	MS33678	Connector, Receptacle, Electric, Integral Mounting

* (Copies of specifications and standards required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

* 2.1.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

OFFICIAL CLASSIFICATION COMMITTEE

Uniform Freight Classification Rules

(Application for copies should be addressed to the Official Classification Committee, Room 202, Union Station, 516 West Jackson Boulevard, Chicago IL 60606.)

* AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D 3951 - Standard Practice for Commercial Packaging.

(Application for copies of ASTM publications should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103.)

2.1.3 Order of Precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

* 3.1 First Article. When specified, a sample shall be subjected to First Article Inspection (see 4.4 and 6.3).

3.2 Materials.

3.2.1 Metals. Metals shall be corrosion-resistant unless protected to resist corrosion during normal service life. The use of dissimilar metals,

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with the parking brake applied and locked. The transmission shall permit grade

especially brass, copper, or steel in intimate metal-to-metal contact with aluminum or aluminum alloy shall be avoided.

3.2.2 Protective treatment. When materials are used in the construction of the transmitter that are subject to deterioration when exposed to climatic and environmental conditions likely to occur during service usage, they shall be protected against such deterioration in a manner that will in no way prevent compliance with the performance requirements of this specification. The use of any protective coating that will crack, chip, or scale with age or extremes of climatic and environmental conditions shall be avoided.

3.2.3 Selection of materials. Specifications and standards for all materials, parts, and Government certification and approval of processes and equipment which are not specifically designated herein and which are necessary for the execution of this specification, shall be selected in accordance with MIL-STD-143, except as provided in the following paragraph.

3.2.4 Standard parts. Standard parts (MS or AN) shall be used wherever they are suitable for the purpose, and shall be identified on the drawing by their part numbers. Commercial utility parts such as screws, bolts, nuts, cotter pins, et cetera, may be used, provided they possess suitable properties and are replaceable by the standard parts (MS or AN) without alteration, and provided the corresponding standard part numbers are referenced in the parts list and, if practicable, on the contractor's drawings. In the event there is no suitable corresponding standard part in effect on date of invitation for bids, commercial parts may be used provided they conform to all requirements of this specification.

3.2.5 Synthetic materials. Synthetic materials shall be of the thermal-setting type composed of a suitable filler and phenol condensation product binder.

3.2.6 Fungusproof materials. Materials which are not nutrients for fungi shall be used to the greatest extent practicable. In cases where materials that are nutrients for fungi must be used, such materials shall be treated with a fungicidal agent, as approved by the Procuring Agency.

3.2.7 Reclaimed materiels. The use of reclaimed materiels shall be encouraged to the maximum extent possible.

3.3 Design. The transmitter shall conform to Figure 1.

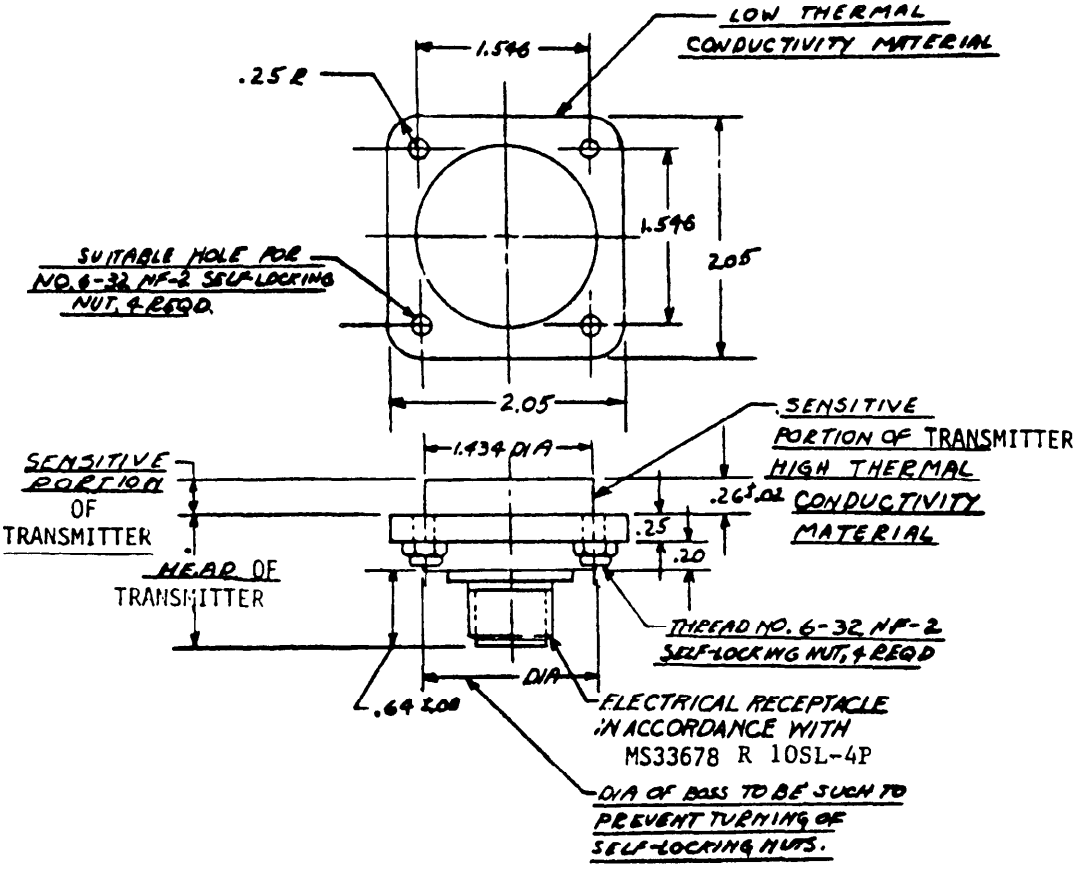
3.4 Construction. The transmitter shall be constructed so that no parts will work loose in service. It shall be built to withstand the strains, jars, vibrations, and other conditions incident to shipping, storage, installation, and service, without failure.

3.5 Interchangeability. All parts having the same manufacturer's part number shall be directly and completely interchangeable with each other with respect to installation and performance. Changes in manufacturer's part numbers shall be governed by the drawing number requirements of DOD-D-1000.

3.6 Finish. All metals shall be suitably protected to meet the requirements of this specification.

3.7 Temperature-resistance characteristics. The temperature-resistance characteristics of the transmitter shall conform within 0.25 percent to the values tabulated in Table I and to all points on straight line interpretations between adjacent tabulated points within the range of -70°C and +70°C.

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DIMENSIONS IN INCHES.
UNLESS OTHERWISE SPECIFIED,
TOLERANCES:
DECIMALS 2 PLACE ±.01
3 PLACE ±.005

FIGURE 1. TRANSMITTER

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TABLE I. TEMPERATURE-RESISTANCE CHARACTERISTICS

°C	Ohms Resistance	°C	Ohms Resistance
-70	797.44	+10	1262.64
-60	851.70	+20	1326.85
-50	906.90	+30	1392.71
-40	963.13	+40	1460.33
-30	1020.47	+50	1529.82
-20	1079.00	+60	1601.30
-10	1138.82	+70	1674.87
0	1200.00		

3.8 Resistance element. The resistance element shall be sealed to withstand immersion of the transmitter in liquid baths, as specified in the tests herein.

3.9 Weight. The weight of the complete transmitter shall not exceed 0.25 pound.

3.10 Screw threads. The threads of all machine screws greater than 0.060 inch in diameter shall conform to MIL-S-7742.

3.11 Gaskets. Three gaskets constructed from black molded gasket rubber in accordance with MIL-R-6855, Class II, of each type, as shown on Figure 2 shall be furnished with each transmitter.

3.12 Mounting lugs and screws. Each mounting lug shall be fitted with a self-locking nut, with No. 6-32 NF-2 threads, securely attached to prevent rotation when the screw is installed. The contractor shall furnish sufficient MS24693 screws for installing the transmitter.

3.13 Identification of product. Equipment, assemblies, and parts shall be marked for identification in accordance with MIL-STD-130.

3.14 Envelope. An envelope, containing the mounting screws and gaskets, shall be furnished with each transmitter. The following information shall be printed on the face of the envelope:

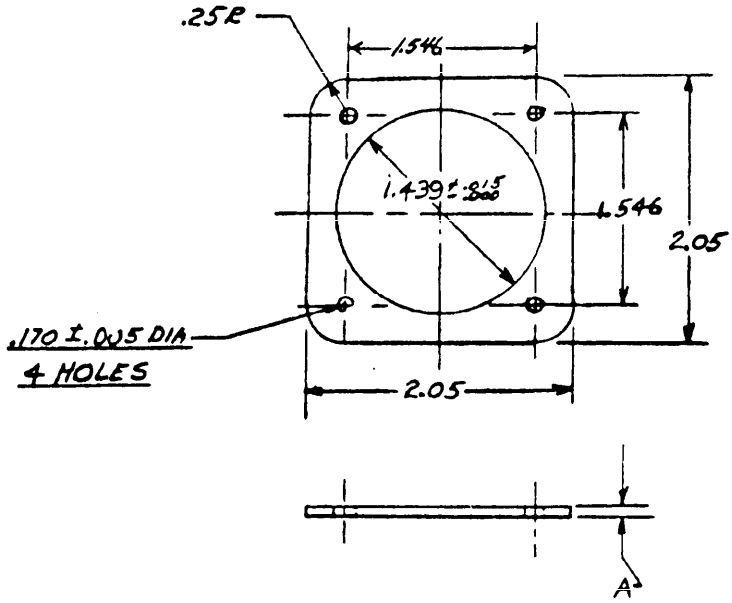
IMPORTANT
THIS ENVELOPE CONTAINS MOUNTING SCREWS AND GASKETS

* 3.15 Workmanship. The transmitter, including all parts and assemblies, shall be constructed and finished in a manner that exemplifies the highest existing instrument production and practices. All surfaces, including threads, shall be free from burrs and sharp edges. All materials shall be sound, of uniform quality and condition, and free from seams, cracks, and other defects which may adversely affect the strength, endurance, usefulness, or wear resistance of the part. Any material which has been treated in any manner to conceal defects therein shall not be offered for government acceptance. All finished surfaces shall be protected against damage and corrosion during manufacture prior to delivery.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the contractor may use his own or any other

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TYPE	A
1	.031 ±.008
2	.062 ±.015
3	.094 ±.015

DIMENSIONS IN INCHES.
UNLESS OTHERWISE SPECIFIED,
TOLERANCES:
DECIMALS 2 PLACE ±.01
3 PLACE ±.005

FIGURE 2. GASKETS

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facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of tests. The inspection and testing of transmitters shall be classified as follows:

* a. First Article Inspection. (See 4.4)

* b. Quality Conformance Inspection. (see 4.5)

4.3 Test conditions.

4.3.1 Standard atmospheric conditions. Whenever the pressure and temperatures existing at the time of the test are not specified definitely, it is understood that the test is to be made at atmospheric pressure (approximately 29.92 inches mercury) and at room temperature (approximately +25°C). When tests are made with atmospheric pressure or room temperature differing materially from the above values, proper allowance shall be made for the difference from the specified condition.

4.3.2 Resistance values. The resistance values given for test results shall be the averages of forward and reversed readings to eliminate thermal-electric potential errors.

4.3.3 Vibration stand. Whenever a vibration stand is specified, it shall vibrate at any desired frequency between 5 to 50 Hertz and shall subject the transmitter to vibration such that a point of the transmitter will describe in a plane inclined 45 degrees to the horizontal plane, a circle, the diameter of which is specified herein.

4.3.4 Sensitive portion. For test purposes, the sensitive portion of the transmitter shall be considered as from the front of the transmitter to the front of the mounting flange, as shown on Figure 1. The head of the transmitter shall be considered as from the front of the mounting flange to the top of the electrical receptacle, as shown on Figure 1.

4.3.5 Temperature-Resistance Measurements. Unless otherwise specified, all temperature-resistance measurements shall be made in agitated liquid bath with the sensitive portion of the transmitter immersed.

4.3.6 Test voltage. All resistance measurements shall be made with 6.0 $\pm$ 0.2 volts direct current (DC) applied to the transmitter.

* 4.4 First Article Inspection.

4.4.1 First Article Tests. First Article Test Samples shall consist of 3 transmitters. First article information in Section 4 shall include the following:

Description of the test routines
Sequence of tests
Number of units to be tested
Data required
Criteria for determining conformance to the requirement

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* 4.4.2 Tests. The first article test shall consist of all the tests specified herein.

4.5 Inspection tests.

4.5.1 Unless otherwise specified, contractor's record of all inspection work and tests, giving the results of tests required to determine compliance with the requirements and tests specified herein, shall be kept complete and shall be available to the Government representative at all times. The record or report of inspection and tests shall be signed or approved by a responsible person specifically assigned by the contractor. Contractors not having laboratory testing facilities satisfactory to the Government shall engage the services of a commercial testing laboratory capable of conducting tests to determine compliance with all the requirements and tests in the specification, and acceptable to the Government.

4.5.2 Individual tests. Each transmitter shall be subjected to the following tests:

4.5.2.1 Examination of product. Each transmitter shall be inspected to determine compliance with the requirement specified herein with respect to material, workmanship, and marking.

4.5.2.2 Transmitter adjustment. The resistance of the transmitter shall be measured at room temperature. The measured resistance shall be within 0.25 percent of the resistance specified in Table I for the exact temperature of the master. To determine the true resistance equivalent to the temperature of the master, interpolation of the values given in Table I may be necessary.

4.5.3 Sampling Plan A tests. Sampling shall be performed in accordance with MIL-STD-105. Inspection Level II, AQL 2.5 percent.

4.5.3.1 Lot. A lot shall consist of transmitters manufactured under essentially the same conditions and submitted for inspection at substantially the same time.

4.5.3.2 Transmitter resistance. The resistance of each transmitter shall be measured at temperatures of -70 ±5°C, 0°C, and +70 ±5°C, in an agitated liquid bath. If desired by the Inspector, resistance measurements at any other points in Table I may also be made. The measured resistance shall be within the tolerances specified in Table I for the exact temperature of the master. To determine the true resistance, equivalent to the temperature of the master, interpolation of the values given in Table I may be necessary. Test records must be maintained by the contractor and made available to the procuring officer on request.

4.5.3.3 Vibration. The transmitter shall be mounted on a suitably designed vibration stand as specified herein. The transmitter shall be subjected to vibration, such that the circle described in the test conditions will have a diameter between 0.030 and 0.32 inch. The vibration frequency shall be varied from 300 to 3000 cpm and return each hour of the test. The transmitter shall be subjected to this test for a 2-hour period. No failure of any kind shall occur during the vibration. To indicate any failure during this test, an indicator or a Wheatstone bridge may be connected to the transmitter. Following the 2-hour vibration period, the transmitter shall be subjected to the transmitter adjustment test. The measured resistance shall be within 0.25 percent of the resistance specified in Table I for the exact temperature of the master.

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4.5.4 Sampling Plan B Tests. Sampling shall be in accordance with MIL-STD-105, Special Inspection Level S-1, AQL 6.5 percent.

4.5.4.1 Detail examination. The transmitter shall be critically examined to determine full compliance with regard to weight, physical dimensions and markings.

4.5.4.2 Response time. The transmitter, with sensitive portion immersed, shall be quenched from an agitated liquid bath at +70°C to an ice bath at 0°C. The transmitter shall be agitated in the ice bath. The time required for the transmitter temperature to fall from +70°C to +10°C shall be noted. The time required for the transmitter temperature to change from +70°C to +10°C shall not exceed 6 seconds.

4.5.4.3 Thermal conductivity. With heated air, the head of the transmitter shall be maintained at a temperature of +70 ±2°C. The sensitive portion of the transmitter shall be subjected to a temperature of -50 ±2°C in an agitated liquid bath. The measured resistance of the transmitter shall not differ by more than 7 ohms from the resistance determined in the transmitter resistance test for the exact temperature of the master.

4.5.4.4 Self heating. The transmitter shall be placed in an agitated liquid bath at a temperature of +25 ±2°C. A current of 40 milliamperes (ma) shall be applied to the transmitter for a minimum period of 2 minutes and the maximum rise in resistance measured during this period. The rise shall not exceed 50 ohms.

4.5.4.5 Thermoelectric potential. The potential produced in the transmitter winding measured with a potentiometer when the sensitive portion of the transmitter is subjected to a temperature of +70 ±2°C in an agitated liquid bath shall not exceed one millivolt.

4.5.4.6 Cold resistance. The transmitter shall be subjected to a temperature of -65°C for a period of 48 hours. With the transmitter at -65°C, it shall be subjected to vibration for a period of 10 minutes. There shall be no damage to the transmitter as a result of this test.

4.5.4.7 High temperature. The transmitter shall be subjected to a temperature of +71°C for a period of 15 hours. With the transmitter at +71°C, it shall be subjected to vibration for a period of 10 minutes. There shall be no damage to the transmitter as a result of this test.

4.5.4.8 Humidity. The transmitter shall be subjected to humidity tests, Method 507.2 of MIL-STD-810. There shall be no rusting or corrosion of the transmitter or any other damage as a result of the test.

4.5.4.8.1 Insulation leakage after humity. A potential of 100 volts DC, shall be applied between any electrical receptacle pin contact and any metal part of the transmitter. Air drying of the transmitter receptacle, connection leads, and outer housing prior to conducting this test is permissible. The insulation resistance measured between any pin and any metal part of the transmitter shall be equal to or greater than 20 megohms.

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4.5.4.9 Transmitter resistance test. Following the completion of the foregoing tests, the transmitter resistance test shall be performed (see paragraph 4.5.3.2).

4.5.4.10 Fungus. A fungus test in accordance with MIL-STD-810, Method 508.3 shall be conducted to determine the transmitter resistance to fungus. Upon completion of the test period, there shall be no signs of fungus growth on the transmitter.

4.5.4.11 Tensile strength. The mounting flange of the transmitter with the sensitive element removed shall be mounted on the movable head of an Amsler testing machine, or equivalent apparatus, with the diameter of the transmitter in a horizontal plane so that the mounting lugs receive no added support. A suitable pin shall be inserted through the hole in the mounting lug of the completely assembled transmitter and shall withstand for one minute, without fracture, a load of 175 pounds applied along the insert nut axis and toward the sensitive portion of the transmitter.

4.5.4.12 Mounting nuts. Each mounting nut shall have a load of 25 pounds applied to the nut in a manner tending to pull the nut out of the lug toward the rear of the transmitter. The mounting nut shall not become loose in the lug and shall not rotate more than a fraction of a turn under a torque of 10 pound-inches.

4.6 All parts, specimens, or assemblies destroyed in making tests required by this specification to determine compliance with the specification shall be in addition to the quantity specified in the contract or purchase order and shall be furnished without increasing the cost of the contract or order.

4.7 Inspection of the preservation, packaging, packing and marking for shipment and storage. Sample items or packs and the inspection of the preservation, packaging, packing and marking for shipment and storage shall be in accordance with the requirements of Section 5, or the documents specified therein.

* 5. PACKAGING

* 5.1 Preservation. Preservation shall be Level A, C or commercial.

* 5.1.1 Level A. Each transmitter shall be preserved and packaged in accordance with MIL-P-116, Method IC1 (no preservative) and placed in a snug fitting container conforming to PPP-B-566.

* 5.1.2 Level C. The Level C preservation for transmitters shall conform to the MIL-STD-794 requirements for this level.

* 5.1.3 Commercial. The commercial preservation of the transmitters shall be in accordance with the requirements of ASTM D 3951.

* 5.2 Packing. Packing shall be Level B, C or Commercial, as specified (see 6.2).

* 5.2.1 Level A. Transmitter, preserved and packaged as specified in 5.1.1, shall be packed in a container conforming to PPP-B-601, overseas type. The closure shall be in accordance with the appendix of the container specification.

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* 5.2.2 Level B. Transmitter preserved and packaged as specified in 5.1.1 shall be packed in weather-resistant shipping containers conforming to PPP-B-636. The grade of each shipping container, determined by the gross weight and size, shall comply to the applicable specification requirements table of the specification. Where practicable, the shipping containers shall be of uniform shape and size and of minimum cube and tare. Closure and strapping shall be in accordance with the appendix of PPP-B-636.

* 5.2.3 Level C. Same as Level A, except that domestic shipping containers may be used.

* 5.2.4 Commercial. The preserved transmitter shall be packed in accordance with ASTM-D-3951.

* 5.3 Marking.

* 5.3.1 Levels A, B, and C. In addition to any special or other identification marking required by the contract (see 6.2), each unit pack, intermediate and exterior container and unitized load shall be marked in accordance with MIL-STD-129.

* 5.3.2 Commercial. Commercial marking shall be in accordance with the requirements of ASTM D 3951.

6. NOTES

6.1 Use. The Type B-4 transmitter covered by this specification is intended for mounting flush on aircraft skins for free air temperature measurement.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number and date of this specification.

b. Levels of packaging and packing (see 5.1 and 5.2).

* 6.3 First Article. When a First Article Inspection is required, the item will be tested and should be a First Article sample or it may be a standard production item from the contractor's current inventory. The First Article should consist of 3 units. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, tests and approval of the First Article.

6.4 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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3.4.7 Seating. With the vehicle on level ground, the [illegible]

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NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions - Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-B-7258B(USAF)	Transmitter, Temperature, Electrical Resistance, Type B-4

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
San Antonio ALC/MMEDO Kelly AFB, TX 78241	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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